



Graph-Based Retrieval-Augmented LLMs for Science



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INTRODUCTION

Quark-gluon-tomography (QGT) Collaboration studies how quarks and gluons are arranged and move inside protons and neutrons, which are the building blocks that make up atomic nuclei. Although protons and neutrons may seem fundamental, they are in fact built from quarks and gluons. Together, quarks and gluons are often called partons. A major goal of modern nuclear physics is to understand how these partons behave, how their momentum is distributed, and how they give rise to the overall structure of matter. Because partons cannot be observed directly, physicists rely on scattering experiments and theoretical models to infer their properties from measurable data.

Retrieval-Augmented Generation, or RAG, is a method that improves how large language models answer questions by allowing them to search for relevant information before generating a response. On their own, language models generate answers based on patterns learned during training, but they do not always have access to the exact documents, datasets, or recent information a user may need. RAG addresses this limitation by combining two steps: retrieval and generation. First, the system searches a database or collection of documents for information related to the user's question. Then, the language model uses that retrieved information to produce a more accurate, grounded, and context-aware answer.

Different RAG architectures retrieve knowledge in different ways. VectorRAG is the standard form of RAG, where document chunks are embedded into vectors and the system retrieves the chunks most similar to the user's query. This works well for direct question answering, but it can struggle when information is spread across multiple documents. GraphRAG adds more structure by building a knowledge graph of entities and relationships from the corpus, allowing the system to answer both local questions about specific concepts and global questions about broader themes in the dataset. HippoRAG goes a step further by using graph-based associative retrieval inspired by human memory (the hippocampus), making it especially useful for reasoning tasks where the answer depends on connecting several related facts. In short, VectorRAG retrieves by similarity, GraphRAG retrieves by structure, and HippoRAG retrieves by association.

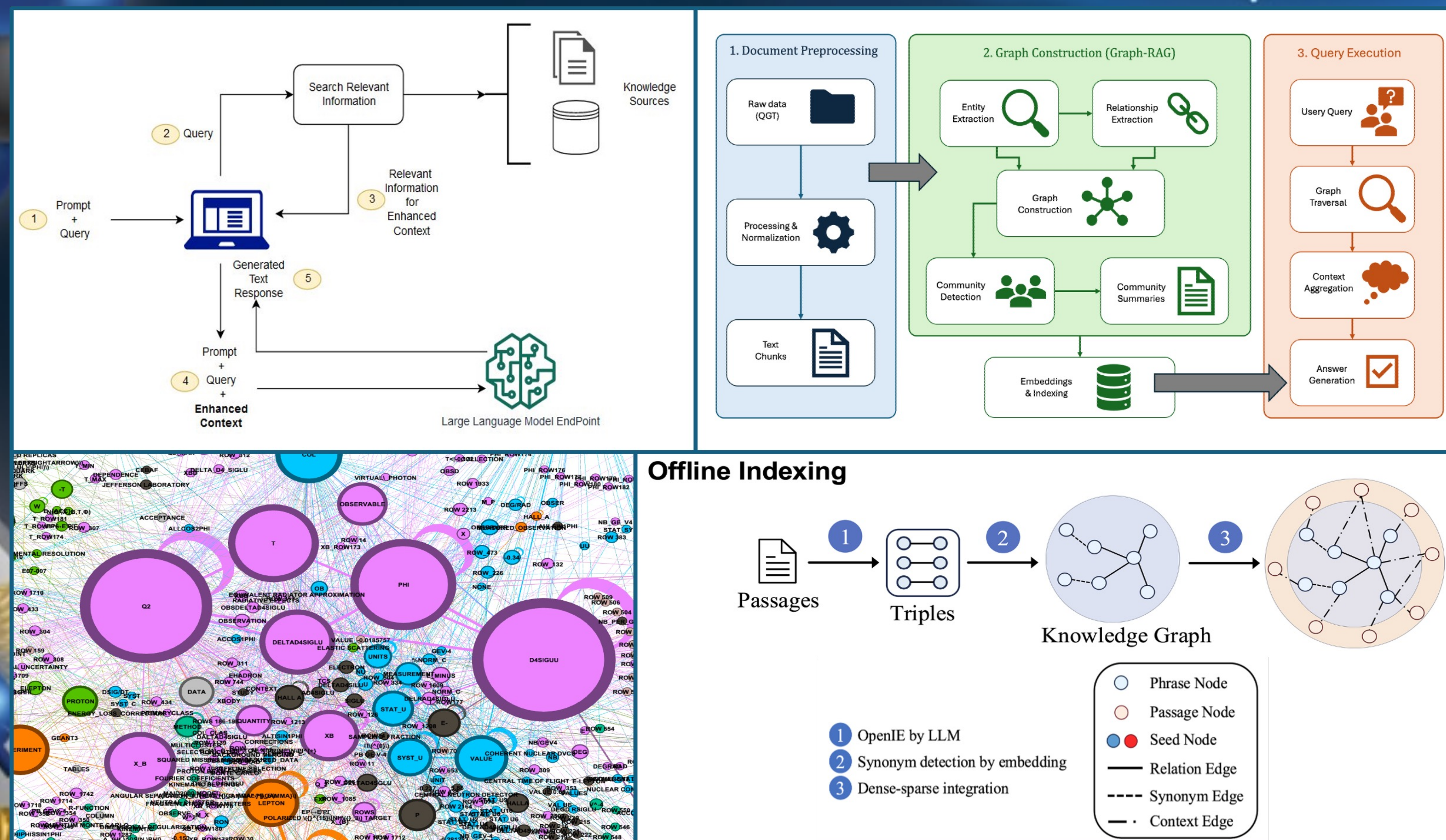
The QGT-database is a repository of QGT-related experimental literature and extracted tabular data, with linked papers, notes, and example notebooks for reading and visualization. RAG can help QGT researchers by focusing LLM outputs in the QGT-database rather than relying on model memory alone. However, QGT data is linking numerical tables, metadata, observables, experiments, and papers, in which a graph-based RAG approach is especially valuable. When combined with an agentic LLM, the system can plan multi-step research tasks, traverse relationships across the database, call analysis tools, verify outputs, and return traceable scientific summaries. In this way, agentic RAG can function as a research assistant that accelerates literature review, data interpretation, and exploratory analysis while preserving scientific context.

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CURRENT STEPS AND PROGRESS

Currently, we are working on improving our Agentic model by crafting an evaluation dataset, consisting of questions relating to pertinent information in the database. The questions range in difficulty and depth, ranging from simple retrieval of information that can be found within the dataset, to tasks that require a deep understanding of multiple papers, cross-referencing mathematical equations and facts to create responses that extrapolate on the information provided in the papers. These questions will provide an important perspective on how the model operates, providing key insight on processes that allow us to diagnose any potential issues early on, such as inconsistent responses, issues with the raw data itself, or limitations within our own code that can all be fixed before the model is published. Ultimately, this evaluation process is essential for validating the model's accuracy, identifying weaknesses, and ensuring that the final system is dependable enough for real scientific use in QGT research.



CONCLUSIONS AND OUTLOOK

RAG, especially when combined with graph-based retrieval and agentic LLM workflows, can make scientific databases much more useful for researchers. Rather than only finding similar text, these systems can connect papers, data tables, metadata, and scientific relationships in a more meaningful way. Our current focus on building an evaluation dataset is helping us test the model carefully and improve it before release. In the long term, this can lead to an AI research assistant that helps scientists explore data, understand papers, and draw connections across the QGT literature more efficiently.

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